What the Iowa data center nitrate study can establish

Research brief

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The reported zero effect is unsupported

The study's nitrate outcomes and weather controls were generated in code, then described as historical water records. Re-executing the scripts reproduced the headline estimates. It did not verify a real-world finding that data centers have no effect on nitrate.

What the audit found

- All 4,176 quarterly nitrate observations were constructed from assumed baselines, seasonal patterns, selected overrides, and random noise. The code did not analyze downloaded nitrate measurements.
- A deliberately extreme input nitrate record left the output unchanged because the purported raw-file ingestion path never read the file.
- The claimed group-time uncertainty was created by multiplying another model's standard errors. Several spatial counts and staggered estimates did not match the written report.
- All 48 water-system IDs matched Iowa's public index. That successful check validates system identities, not the nitrate measurements or facility-to-water-source links.

Table 1 The headline calculations

Generated-data estimate	Reported change
Construction within 10 km	+0.0013 mg/L as N
Operation within 10 km	−0.0124 mg/L as N

These calculations reproduce from generated data. They are not estimates of actual changes in Iowa drinking water.

What this means for policy

This project establishes neither harm nor absence of harm from data centers. Its zero-effect, equivalence, and weather-attribution claims should be removed from public communications. Water demand, treatment capacity, raw-water contamination, and finished-water concentration are separate outcomes that require separate evidence.

Nitrate-treatment constraints during the 2025 Central Iowa Water Works conservation response are documented by the utility. That context does not establish which industries or land uses caused the measured nitrate concentrations.

Sources Repository snapshot 7e1f0e6; original manuscript and Python scripts; Iowa DNR Source Water Protection Tracker; Central Iowa Water Works Stage III Daily Updates. Detailed evidence and executable checks are in the accompanying peer review and code package.

Actual public records provide a starting point

The replacement workflow downloaded EPA nitrate records for 2012–2019 and preserved observed values, sample-specific detection limits, missing periods, and unresolved water-stage labels.

Available observed data	Coverage
Iowa nitrate records	28,103
Usable routine community samples	17,011 from 845 systems
Unresolved raw or finished stage	16,489 samples or 96.9%
Finished or entry stage	521 samples from 70 systems
Usable system-quarter coverage	9,866 of 27,040 cells or 36.5%
Original 48 systems present in EPA release	38 systems with 2,355 usable samples

System-quarter denominator is limited to community systems appearing in the EPA release. Missing records do not establish a utility's failure to monitor. These data end in 2019 and cannot test claims about 2025–2026.

What a credible follow up requires

- Obtain dated raw and finished nitrate samples, sampling locations, method-specific reporting limits, and treatment or supplier changes through the study end date.
- Verify actual land-disturbance and cooling-withdrawal dates, separately for campuses and expansions. Approval dates and planned projects cannot substitute for operation.
- Connect facilities to real intake catchments, groundwater capture zones, or municipal suppliers. City names and distance alone are insufficient.
- Use comparable unexposed water sources, measured weather, and enough independent treated supply systems for credible inference. Preserve unobserved periods and report sensitivity to censoring and sampling intensity.

What has been delivered

The package includes the repository snapshot, forensic replication outputs clearly marked as generated, the observed EPA extract, scripts for extraction and cleaning, coverage analyses, a Colab notebook, verification checks, a fresh manuscript, and an independent peer-review report. The causal-analysis path stops until reviewed exposure and source links are supplied.

Primary data EPA Six-Year Review 4 Compliance Monitoring Data 2012–2019 and its user guide.
https://www.epa.gov/dwsixyearreview/six-year-review-4-compliance-monitoring-data-2012-2019